SECURE VISION-LANGUAGE MODELS IN BIG DATA ANALYTICS

Comparative Analysis: Healthcare and Agriculture Applications

Group 20

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INTRODUCTION

Vision-language models is a fusion of **vision and natural language foundational model,** has enhanced contextual understanding.

With the increased usage of data, **data security and privacy** consideration has seen an elevated emphasis.

Integrating both key elements, the chosen two informative articles :

A1

Federated vision-language models for **medical diagnosis**

A2

Lightweight federated visual-language detection for **smart farming**

PROBLEM STATEMENT

Data Privacy

A1: Medical institutions unable to share patients information

A2: Farmland data concern when centralised

Unimodal Limitation

A1: Text or image only provide incomplete diagnostic information

A2: Traditional object detection model lacks semantic understanding

Overhead Comms.

A1: Repeated transmission of weights (172 params), cause bandwidth strain

A2: Rural farm networks with transmission of high params, causes delay

METHODOLOGIES

HEALTHCARE | AGRICULTURE

SECURE MULTIMODAL PIPELINE

Privacy-Preserving Federated Learning Pipeline

- **Collection:** Multimodal data (X-rays + Clinical Reports) is collected at local hospital nodes (clients).
- **Transformation:** Data is pre-processed locally where raw patient data never leaves the hospital firewall.
- **Storage:** Decentralized storage where only model updates are aggregated globally.

Secure Edge-Hybrid Pipeline

- **Collection:** UAVs and IoT sensors collect high-resolution imagery and field data.
- **Transformation:** Edge devices perform noise reduction and lightweight preprocessing to reduce bandwidth.
- **Storage:** Processed data is encrypted and transmitted to a centralized cloud for heavy analytics.

NEXT-GEN FOUNDATION MODELS

	Base Model	Text Encoder	Vision Encoder	Vision-LLM
Healthcare	GroundingDINO	BERT	Swin Transformer	Llama-3.2
	172M Parameter	Extract text from medical report	Extract medical image feature	Generate medical report
Agriculture	GroundingDINO	BERT	Swin Transformer	
	172M Parameter	Class embeddings	Crop image analysis	

HYBRID ANALYTICAL MODELS & TECHNIQUES

Low-Rank Adaptation (LoRA)

- **Technique:** Freezes the massive pre-trained Llama model and injects trainable rank decomposition matrices.
- **Benefit:** Allows efficient fine-tuning on hospital GPUs without high computational cost.

Zero-Shot Learning & Edge Inference

- **Technique:** The model can identify unseen pest classes without retraining.
- **Benefit:** Real-time inferencing on edge devices allows for immediate action in the field.

SECURITY MEASURES

Differential Privacy (DP) & Secure Aggregation

- **Mechanism**: Adds statistical noise to model gradients before sharing them.
- **Goal:** Prevents attackers from reverse-engineering specific patient data from the model updates.

AES-256 Encryption & Access Control

- **Mechanism:** End-to-end encryption for data in transit.
- **Goal:** Protects proprietary crop data and prevents unauthorized geolocation tracking.

References *

- Li, L., Li, J., Chen, D., Pu, L., Yao, H., & Huang, Y. (2025). VLLFL: A Vision-Language Model Based Lightweight Federated Learning Framework for Smart Agriculture. arXiv preprint arXiv:2504.13365.
- Wang, N., Deng, Y., Fan, S., Yin, J., & Ng, S. K. (2025). Multi-Modal One-Shot Federated Ensemble Learning for Medical Data with Vision Large Language Model. arXiv preprint arXiv:2501.03292.

* *Based on preprints as there was limitation is finding peer-reviewed articles that address both data privacy and multimodal foundation model*

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STRENGTHS

ARTICLE 1 (FedMME)

Scalability: Deployed successfully across 5-29 medical institutions.

Accuracy: Improvement 17.5% on RSNA dataset & 87.93% mAP on Blood dataset.

Flexibility: Adapts to various medical imaging modalities(CT, X-ray, retinal scans)

Privacy: One-shot learning eliminates multiple data exposure rounds

Multimodal advantage: Combining images + reports outperforms image-only approaches

Robustness: Performs well under high non-IID conditions (a=0.1)

ARTICLE 2 (VLLFL)

Scalability: Generalizes across fruit detection and harmful animal detection tasks

Accuracy: 14.55% mAP improvement (9.95% → 24.5%)

Communication efficiency: 99.3% reduction in transmitted data (656 MB → 3.81 MB)

Flexibility: Context-aware prompts adapt to diverse agricultural scenarios

Fast convergence: Achieves significant accuracy after just 10 training epochs

Generalization: Works across different crops, fruits, and animal species

WEAKNESSES

High computational cost: Requires significant GPU resources for vision-LLM inference

Dataset requirements: Needs sufficient medical images per institution for effective training

Complexity: Multi-stage pipeline increases implementation difficulty

Limited by base model: Performance ceiling constrained by GroundingDINO capabilities

Text modality dependency: Generated reports quality affects overall performance

Hardware constraints: Vision-LLM deployment challenging in resource-limited hospitals

High computational cost: GroundingDINO requires substantial inference resources

Data privacy concerns: Prompt leakage could potentially reveal class distribution information

Implementation complexity: Requires careful prompt width tuning and embedding alignment

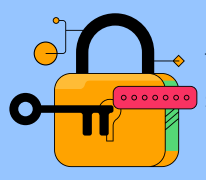
Performance ceiling: Cannot exceed base model maximum achievable accuracy

Domain specificity: Requires retraining prompt generator for substantially different tasks

Network dependency: Federated learning requires stable connectivity across farms

KEY FINDINGS

Model-Aware Security
Security mechanisms are most effective when designed in alignment with the capabilities and constraints of multimodal foundation models rather than applied as external layers.



Expanded attack surface
The inclusion of text, images, and structured clinical data increases security and privacy risks, particularly from image data, making secure preprocessing and modality-specific validation essential.

Security vs. Efficiency Trade-Off
A trade-off exists between strong security and efficiency. Article 1 prioritizes robust privacy controls, whereas Article 2 emphasizes scalability and real-time performance with acceptable security.

BEST PRACTICE

End-to-end protection
Collection, Processing, Storage & Inference

Modality-Specific Controls
Text, Images & Structured Data Safeguards

Context-Driven Modal Choices
Select models based on security needs

Auditing & Compliance
Logging and regulatory readiness

FUTURE DIRECTIONS

	Future focus: Healthcare AI	Future focus: Smart Agriculture	Future focus: Healthcare AI	Future focus: Smart Agriculture
Technology features	Designed **advanced algorithms with privacy** for sensitive language medical models	**Innovating with visual feature prompts** for better task performance	Build **expanding systems for real-time diagnostics** support	**Expanding to crop and livestock management** with application of AI
Model pipelines	**Develop standardized pipelines** to process diversity of complex medical data	Improve model resillience by **building robustness for field conditions**	Deploy efficiency in **optimizing for hospital environments**	**Enabling in-field edge computing** on devices optimized for rural networks
			Applications	Computational efficiency

References *
- Li, L., Li, J., Chen, D., Pu, L., Yao, H., & Huang, Y. (2025). VLLFL: A Vision-Language Model Based Lightweight Federated Learning Framework for Smart Agriculture. arXiv preprint arXiv:2504.13365.
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